

Coxeter numbers and spectral gaps of rank-three weighted group-algebra Laplacians

Haoran Zhu*

Abstract

We construct a positive weight system on the reflections of the Coxeter group B_3 for which the spectral gap of the corresponding weighted group-algebra Laplacian is attained in a three-dimensional irreducible representation of generalised Coxeter number 8, although the Coxeter number of B_3 is 6. In particular, the spectral gap is not always attained in an irreducible representation of generalised Coxeter number equal to the Coxeter number. This gives a negative answer to the first part of Question 9.3 of Chapuy and Douvropoulos.

Keywords: complex reflection groups; weighted group-algebra Laplacians; spectral gap; generalised Coxeter number.

Mathematics Subject Classification: 20F55; 20C15; 05C81.

1 Introduction

Let W be a finite complex reflection group with reflection set $\mathcal{R}$, and let $w : \mathcal{R} \rightarrow \mathbb{R}_{>0}$ be a positive weight system satisfying $w(\tau) = w(\tau^{-1})$ for all $\tau \in \mathcal{R}$. Following Chapuy and Douvropoulos [6, Definition 3.12], we define the associated **group-algebra Laplacian** by

$$\mathcal{L}_W(w) := \sum_{\tau \in \mathcal{R}} w(\tau) (1 - \tau) \in \mathbb{C}[W]. \quad (1.1)$$

*Division of Mathematical Sciences, Nanyang Technological University, 21 Nanyang Link, Singapore 637371.
Email: zhuh0031@e.ntu.edu.sg

These operators appear naturally in the weighted reflection-factorisation formulas and matrix-tree theorems of Chapuy and Douvropoulos [6]. Acting in the left-regular representation, $\mathcal{L}_W(w)$ is the unnormalised Laplacian of the weighted Cayley graph of W with generating set $\mathcal{R}$. Its smallest positive eigenvalue is the spectral gap, and the resulting problem is to determine which irreducible representations can realise it.

For $W = \mathfrak{S}_n$, the situation is much more rigid. The theorem of Caputo, Liggett and Richthammer [2], which proves Aldous' spectral gap conjecture, implies that, for arbitrary positive weights on transpositions, the spectral gap is realised in the standard representation. In the present group-algebra language, this reformulation is explained in [4, Section 3]; see also [5] for related remarks. Related Aldous-type results and variants may be found in [1, 3, 8, 9, 10].

For a general well-generated irreducible complex reflection group, the unweighted element

$$\mathcal{L}_W := \sum_{\tau \in \mathcal{R}} (1 - \tau)$$

acts on each irreducible representation by the corresponding generalised Coxeter number in the sense of Gordon and Griffeth [7, §1.3]. In particular, the usual Coxeter number h is the smallest positive generalised Coxeter number occurring among non-trivial irreducible representations [6, Proposition 3.13, Definition 3.14 and §9.3].

This suggests the following analogue of Aldous' problem: for an arbitrary positive weight system on the reflections of W , must the spectral gap of $\mathcal{L}_W(w)$ be attained in an irreducible representation of generalised Coxeter number h ? This is exactly the first part of Question 9.3 of Chapuy and Douvropoulos [6]. The dihedral example discussed in [6, §9.3] does not settle the present question, because there the competing irreducible representation still has generalised Coxeter number h .

The purpose of this note is to show that the answer is negative by giving a rank-three counterexample.

Theorem 1.1. *There exists a positive weight system w on the reflections of B_3 such that*

$$\lambda_{\text{gap}}(\mathcal{L}_{B_3}(w)) = 17 - \sqrt{33},$$

and this eigenvalue is attained in a three-dimensional irreducible representation of generalised Coxeter number 8. In particular, the spectral gap is not attained in any irreducible representation of generalised Coxeter number 6.

The paper is organised as follows. In Section 2 we recall the necessary definitions and fix explicit models for the irreducible representations of B_3 . In Section 3 we introduce the weight system, compute the representation Laplacians in every irreducible block, and prove Theorem 1.1.

2 Preliminaries

2.1 Weighted group-algebra Laplacians and spectral gaps

Let $\mathcal{L}_W(w) \in \mathbb{C}[W]$ be the group-algebra Laplacian defined in (1.1). For a finite-dimensional representation $\rho : W \rightarrow \text{GL}(U)$, we define

$$L_\rho(w) := \rho(\mathcal{L}_W(w)) = \sum_{\tau \in \mathcal{R}} w(\tau)(I_U - \rho(\tau)) \in \text{End}(U). \quad (2.1)$$

We call $L_\rho(w)$ the corresponding representation Laplacian.

Since W is finite, ρ may be realised unitarily. With this choice,

$$L_\rho(w)^* = \sum_{\tau \in \mathcal{R}} w(\tau)(I_U - \rho(\tau)^*) = \sum_{\tau \in \mathcal{R}} w(\tau)(I_U - \rho(\tau^{-1})) = \sum_{\tau \in \mathcal{R}} w(\tau^{-1})(I_U - \rho(\tau)) = L_\rho(w),$$

so $L_\rho(w)$ is Hermitian, and in particular has real eigenvalues. Moreover, for every $v \in U$,

$$\begin{aligned} \langle v, L_\rho(w)v \rangle &= \sum_{\tau \in \mathcal{R}} w(\tau)(\|v\|^2 - \langle v, \rho(\tau)v \rangle) \\ &= \frac{1}{2} \sum_{\tau \in \mathcal{R}} w(\tau) \left(2\|v\|^2 - \langle v, \rho(\tau)v \rangle - \langle v, \rho(\tau^{-1})v \rangle \right) \\ &= \frac{1}{2} \sum_{\tau \in \mathcal{R}} w(\tau) \|v - \rho(\tau)v\|^2 \geq 0, \end{aligned}$$

and therefore $L_\rho(w)$ is positive semidefinite.

Let $\lambda : W \rightarrow \text{GL}(\mathbb{C}[W])$ denote the left-regular representation. We define the spectral gap of $\mathcal{L}_W(w)$, denoted $\lambda_{\text{gap}}(\mathcal{L}_W(w))$, to be the smallest positive eigenvalue of $L_\lambda(w)$.

Lemma 2.1. *Let $\widehat{W}$ be a complete set of representatives of the irreducible complex representations of W . Then, with multiplicities, the spectrum of $L_\lambda(w)$ is the union*

of the spectra of the operators $L_\rho(w)$, where the spectrum of $L_\rho(w)$ is repeated $\dim(\rho)$ times for each $\rho \in \widehat{W}$. Consequently,

$$\lambda_{\text{gap}}(\mathcal{L}_W(w)) = \min_{\rho \in \widehat{W} \setminus \{\mathbf{1}\}} \lambda_{\min}(L_\rho(w)),$$

where $\lambda_{\min}$ denotes the smallest eigenvalue.

Proof. By Maschke's theorem,

$$\mathbb{C}[W] \cong \bigoplus_{\rho \in \widehat{W}} U_\rho^{\oplus \dim(\rho)}$$

as a W -module. Since $\mathcal{L}_W(w)$ acts on $\mathbb{C}[W]$ by left multiplication, it follows that

$$L_\lambda(w) \cong \bigoplus_{\rho \in \widehat{W}} L_\rho(w)^{\oplus \dim(\rho)}.$$

This gives the spectral decomposition.

If 0 were an eigenvalue of $L_\rho(w)$ for some non-trivial irreducible representation ρ , then, since $L_\rho(w)$ is positive semidefinite, there would exist $0 \neq v \in U_\rho$ such that

$$0 = \langle v, L_\rho(w)v \rangle = \frac{1}{2} \sum_{\tau \in \mathcal{R}} w(\tau) \|v - \rho(\tau)v\|^2.$$

Hence $\rho(\tau)v = v$ for every $\tau \in \mathcal{R}$. Since the reflections generate W , this makes v a nonzero W -fixed vector, contradicting the irreducibility of ρ unless $\rho = \mathbf{1}$. Therefore 0 occurs only in the trivial block, and the formula for $\lambda_{\text{gap}}(\mathcal{L}_W(w))$ follows. $\square$

2.2 Generalised Coxeter numbers

Assume now that W is a well-generated irreducible complex reflection group of rank n . Consider the unweighted element

$$\mathcal{L}_W := \sum_{\tau \in \mathcal{R}} (1 - \tau) \in \mathbb{C}[W]. \tag{2.2}$$

Since $\mathcal{R}$ is closed under conjugation, $\mathcal{L}_W$ is central in $\mathbb{C}[W]$. For an irreducible character χ , write

$$\chi(\mathcal{R}) := \sum_{\tau \in \mathcal{R}} \chi(\tau).$$

Following Gordon–Griffeth [7, §1.3] and Chapuy–Douvropoulos [6, Definition 3.14], we define the generalised Coxeter number of χ by

$$c_\chi := |\mathcal{R}| - \frac{\chi(\mathcal{R})}{\chi(1)}. \quad (2.3)$$

Equivalently, c_χ is the scalar by which $\mathcal{L}_W$ acts in the irreducible representation with character χ . If ρ is irreducible, we also write c_ρ for this number.

If V_{ref} denotes the reflection representation of W , then $c_{V_{\text{ref}}} = h$, the ordinary Coxeter number of W . Moreover, for well-generated groups, h is the smallest positive generalised Coxeter number occurring among non-trivial irreducible representations [6, Proposition 3.13, Definition 3.14 and §9.3].

2.3 The Coxeter group B_3 and irreducible representations

We realise B_3 as the group of signed permutation matrices in $\text{GL}_3(\mathbb{R})$, with standard basis e_1, e_2, e_3 . It has order 48 and nine reflections. For $i = 1, 2, 3$, let s_i be the short reflection sending $e_i \mapsto -e_i$ and fixing the other basis vectors. For $1 \leq i < j \leq 3$, let t_{ij} be the reflection swapping e_i and e_j , and let u_{ij} be the reflection sending $e_i \mapsto -e_j$ and $e_j \mapsto -e_i$, again fixing the remaining basis vector.

The four linear characters are $\mathbf{1}$, sgn , δ , and $\det := \text{sgn}\delta$. Here sgn is the sign of the underlying permutation, so $\text{sgn}(s_i) = 1$ and $\text{sgn}(t_{ij}) = \text{sgn}(u_{ij}) = -1$, while δ is the sign-change character, so $\delta(s_i) = -1$ and $\delta(t_{ij}) = \delta(u_{ij}) = 1$. Thus $\det(\tau) = -1$ for every reflection $\tau \in \mathcal{R}$.

Let V be the standard three-dimensional reflection representation of B_3 . For each linear character χ , the twist $V \otimes \chi$ is irreducible; in particular, we set $U := V \otimes \delta$. Let $\pi : B_3 \rightarrow \mathfrak{S}_3$ be the quotient obtained by forgetting signs, and let E be the pullback along π of the standard two-dimensional irreducible representation of $\mathfrak{S}_3$. Then E is irreducible, and $E \otimes \delta$ is the other irreducible representation of dimension 2.

Since all short reflections are conjugate in B_3 , and all long reflections are conjugate, it suffices to record the character values on s_1 and on t_{12} . The relevant values and the corresponding generalised Coxeter numbers are:

ρ	$\chi_\rho(s_1)$	$\chi_\rho(t_{12})$	c_ρ
$\mathbf{1}$	1	1	0
δ	-1	1	6
sgn	1	-1	12
$\det$	-1	-1	18
E	2	0	6
$E \otimes \delta$	-2	0	12
V	1	1	6
$U = V \otimes \delta$	-1	1	8
$V \otimes \text{sgn}$	1	-1	10
$V \otimes \det$	-1	-1	12

Indeed, each reflection has trace 1 in V , short reflections act trivially on E , and long reflections act on E as transpositions in the standard representation of $\mathfrak{S}_3$, hence with trace 0. The remaining rows are obtained by twisting with the linear characters, and c_ρ is then computed from (2.3) using $|\mathcal{R}| = 9$.

In particular,

$$c_V = 6 = h, \quad c_U = 8. \quad (2.4)$$

All ten representations listed above are pairwise non-isomorphic. Since

$$4 \cdot 1^2 + 2 \cdot 2^2 + 4 \cdot 3^2 = 48 = |B_3|,$$

this list is complete. In particular, the irreducible representations of generalised Coxeter number 6 are precisely δ , E , and V .

3 A counterexample

We now prove Theorem 1.1 by an explicit computation of the spectra of the operators $L_\rho(w_0)$.

Let $w_0 : \mathcal{R} \rightarrow \mathbb{R}_{>0}$ be the weight system on B_3 given by

τ	s_1	s_2	s_3	t_{12}	u_{12}	t_{13}	u_{13}	t_{23}	u_{23}
$w_0(\tau)$	1	1	4	3	4	1	3	1	3

Since every reflection of B_3 is an involution, the relation $w_0(\tau) = w_0(\tau^{-1})$ is automatic. Write

$$S := w_0(s_1) + w_0(s_2) + w_0(s_3) = 6, \quad L := \sum_{1 \leq i < j \leq 3} (w_0(t_{ij}) + w_0(u_{ij})) = 15, \quad (3.1)$$

so that

$$\sum_{\tau \in \mathcal{R}} w_0(\tau) = S + L = 21.$$

We first compute the two three-dimensional representations that drive the argument.

Lemma 3.1. *In the basis (e_1, e_2, e_3) , the representation Laplacians of V and $U = V \otimes \delta$ are*

$$L_V(w_0) = \begin{pmatrix} 13 & 1 & 2 \\ 1 & 13 & 2 \\ 2 & 2 & 16 \end{pmatrix} \quad \text{and} \quad L_U(w_0) = \begin{pmatrix} 21 & 1 & 2 \\ 1 & 21 & 2 \\ 2 & 2 & 12 \end{pmatrix}. \quad (3.2)$$

Consequently,

$$\text{Spec}(L_V(w_0)) = \{12, 12, 18\} \quad (3.3)$$

and

$$\text{Spec}(L_U(w_0)) = \{20, 17 - \sqrt{33}, 17 + \sqrt{33}\}. \quad (3.4)$$

In particular,

$$\lambda_{\min}(L_U(w_0)) = 17 - \sqrt{33} < 12 = \lambda_{\min}(L_V(w_0)). \quad (3.5)$$

Proof. A direct calculation from (2.1) gives (3.2). Their characteristic polynomials are $(x - 18)(x - 12)^2$ and $(x - 20)(x^2 - 34x + 256)$, respectively, which yields (3.3) and (3.4). The final inequality is immediate. $\square$

Thus, the representation U of generalised Coxeter number 8 already has a smaller lowest eigenvalue than the representation V of generalised Coxeter number 6. It remains to rule out the other irreducible representations.

Proposition 3.2. For the weight system above, the spectra of the representation Laplacians in the remaining irreducible representations are

$$\begin{aligned} \text{Spec}(L_1(w_0)) &= \{0\}, & \text{Spec}(L_\delta(w_0)) &= \{12\}, \\ \text{Spec}(L_{\text{sgn}}(w_0)) &= \{30\}, & \text{Spec}(L_{\det}(w_0)) &= \{42\}, \\ \text{Spec}(L_E(w_0)) &= \{12, 18\}, & \text{Spec}(L_{E \otimes \delta}(w_0)) &= \{24, 30\}, \\ \text{Spec}(L_{V \otimes \text{sgn}}(w_0)) &= \{22, 25 - \sqrt{33}, 25 + \sqrt{33}\}, \\ \text{Spec}(L_{V \otimes \det}(w_0)) &= \{24, 30, 30\}. \end{aligned}$$

In particular, together with (3.3) and (3.4), every non-trivial irreducible representation other than U has smallest Laplacian eigenvalue at least 12.

Proof. For a linear character χ ,

$$L_\chi(w_0) = \sum_{\tau \in \mathcal{R}} w_0(\tau)(1 - \chi(\tau)).$$

Using the values of $\mathbf{1}$, δ , sgn , and $\det$ on short and long reflections, together with (3.1), one obtains

$$L_{\mathbf{1}}(w_0) = 0, \quad L_\delta(w_0) = 12, \quad L_{\text{sgn}}(w_0) = 30, \quad L_{\det}(w_0) = 42.$$

Now consider E . Since E factors through the quotient $\pi : B_3 \rightarrow \mathfrak{S}_3$, each short reflection acts trivially, while t_{ij} and u_{ij} both act as the transposition (ij) . Hence $L_E(w_0)$ depends only on

$$\begin{aligned} \alpha_{12} &:= w_0(t_{12}) + w_0(u_{12}) = 7, \\ \alpha_{13} &:= w_0(t_{13}) + w_0(u_{13}) = 4, \\ \alpha_{23} &:= w_0(t_{23}) + w_0(u_{23}) = 4. \end{aligned}$$

Let P be the permutation representation of $\mathfrak{S}_3$ on $\mathbb{C}^3$. Since $P = \mathbf{1} \oplus E$, the corresponding weighted transposition Laplacian on P is

$$L(\alpha) = \alpha_{12}(I - (12)) + \alpha_{13}(I - (13)) + \alpha_{23}(I - (23)) = \begin{pmatrix} 11 & -7 & -4 \\ -7 & 11 & -4 \\ -4 & -4 & 8 \end{pmatrix}.$$

Its characteristic polynomial is $x(x - 12)(x - 18)$. The zero eigenvalue belongs to the trivial summand, so

$$\text{Spec}(L_E(w_0)) = \{12, 18\}.$$

Twisting by δ changes only the short reflections, each of which acts trivially on E , and therefore

$$L_{E \otimes \delta}(w_0) = L_E(w_0) + 2S I_2 = L_E(w_0) + 12I_2.$$

Hence, $\text{Spec}(L_{E \otimes \delta}(w_0)) = \{24, 30\}$.

For $V \otimes \text{sgn}$, the short reflections contribute exactly as in V , whereas each long-reflection term $I - \rho_V(\tau)$ is replaced by $I + \rho_V(\tau)$. Thus

$$L_{V \otimes \text{sgn}}(w_0) = \begin{pmatrix} 21 & -1 & -2 \\ -1 & 21 & -2 \\ -2 & -2 & 30 \end{pmatrix},$$

whose characteristic polynomial is $(x - 22)(x^2 - 50x + 592)$. Therefore

$$\text{Spec}(L_{V \otimes \text{sgn}}(w_0)) = \{22, 25 - \sqrt{33}, 25 + \sqrt{33}\}.$$

Finally, since $\det(\tau) = -1$ for every reflection $\tau \in \mathcal{R}$,

$$L_{V \otimes \det}(w_0) = 2(S + L)I_3 - L_V(w_0) = 42I_3 - L_V(w_0).$$

Using (3.3), we obtain

$$\text{Spec}(L_{V \otimes \det}(w_0)) = \{24, 30, 30\}.$$

This proves the proposition. □

We can now prove the main theorem.

Proof of Theorem 1.1. By Lemma 3.1 and Proposition 3.2, the smallest eigenvalue among the non-trivial irreducible blocks is $17 - \sqrt{33}$, and it occurs in $U = V \otimes \delta$. Lemma 2.1 therefore gives

$$\lambda_{\text{gap}}(\mathcal{L}_{B_3}(w_0)) = 17 - \sqrt{33}.$$

By (2.4), the representation U has generalised Coxeter number $c_U = 8$, whereas the Coxeter number of B_3 is $h = 6$. Hence the spectral gap is not attained in any irreducible representation of generalised Coxeter number h . □

References

- [1] G. Alon and D. Puder, *Aldous-type spectral gaps in unitary groups*, preprint, arXiv:2603.00353.
- [2] P. Caputo, T. M. Liggett, and T. Richthammer, *Proof of Aldous' spectral gap conjecture*, J. Amer. Math. Soc. **23** (2010), no. 3, 831–851.
- [3] F. Cesi, *Cayley graphs on the symmetric group generated by initial reversals have unit spectral gap*, Electron. J. Combin. **16** (2009), no. 1, Paper N29.
- [4] F. Cesi, *On the eigenvalues of Cayley graphs on the symmetric group generated by a complete multipartite set of transpositions*, J. Algebraic Combin. **32** (2010), no. 2, 155–185.

- [5] F. Cesi, *A few remarks on the octopus inequality and Aldous' spectral gap conjecture*, Comm. Algebra **44** (2016), no. 1, 279–302.
- [6] G. Chapuy and T. Douvropoulos, *Coxeter factorizations with generalized Jucys–Murphy weights and Matrix-Tree theorems for reflection groups*, Proc. Lond. Math. Soc. (3) **126** (2023), no. 1, 129–191.
- [7] I. G. Gordon and S. Griffeth, *Catalan numbers for complex reflection groups*, Amer. J. Math. **134** (2012), no. 6, 1491–1502.
- [8] G. Greaves and H. Zhu, *Aldous property for full-flag Johnson graphs*, preprint, arXiv:2603.10406.
- [9] Y. Li, B. Xia, and S. Zhou, *Aldous' spectral gap property for normal Cayley graphs on symmetric groups*, European J. Combin. **110** (2023), Paper No. 103657.
- [10] O. Parzanchevski and D. Puder, *Aldous's spectral gap conjecture for normal sets*, Trans. Amer. Math. Soc. **373** (2020), no. 10, 7067–7086.